Answer Key - Mathematics - K-12 Level Assessment 20 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 76; B) 77; C) 78; D) 79
   Note: The correct answer is 78 because the average score is calculated by summing the test scores (76 + 80 + 83 + 71 + 80 + 78 = 468) and dividing by the number of tests (6), resulting in an average of 78 (468 ÷ 6 = 78).

2. **Answer:** D
   Options: A) 8.86; B) 42.25; C) 32.54; D) 38.44
   Note: The correct answer of 38.44 is obtained by cross-multiplying the given rates to find the equivalent value of 18.5 dollars per m gallons, leading to the equation 18.5 m = 3.60 * 7.5, which simplifies to m = 38.44 when calculated.

3. **Answer:** D
   Options: A) 12; B) 22; C) 840; D) 240
   Note: The smallest positive integer with factors of 16, 15, and 12 is 240 because it is the least common multiple (LCM) of these numbers, which is found by taking the highest powers of their prime factors: 2^4 from 16, 3^1 from 15, and 5^1 from 15, resulting in $2^4 \cdot 3^1 \cdot 5^1 = 240$.

4. **Answer:** A
   Options: A) 10; B) 16; C) 24; D) 12
   Note: The correct answer is 10 because the given circle can be rewritten in standard form as $(x - 12)^2 + (y + 5)^2 = 25$, indicating a center at (12, -5) with a radius of 5, and the shortest distance from the origin (0, 0) to the circle is the distance from the origin to the center (approximately 13) minus the radius (5), resulting in 13 - 5 = 8.

5. **Answer:** D
   Options: A) -4; B) -2; C) 10; D) 2
   Note: To find the value of -2(x - 3) when x = 2, substitute 2 into the expression to get -2(2 - 3) = -2(-1) = 2, demonstrating the application of substitution and simplification.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because Jane's mean quiz score of 94.5 would require the total of her scores divided by the number of quizzes to equal 94.5, which is not necessarily true if her actual scores do not add up to 472.5.

7. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because if the ratio of 2 x - y to x + y is 2 to 3, it can be manipulated to show that the ratio of x to y simplifies to 5 to 4.

8. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because the interval [-1, 1] does not accurately represent all values of u for which neither 2 u nor -20 u falls within (-, -1), as those inequalities require a different set of conditions for u.

9. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because $\log_3 81$ equals 4, since $3^4 = 81$, not -1.

10. **Answer:** True
    Options: A) True; B) False
    Note: The statement is true because if 3 cans cost $2.37, then the cost per can is 0.79$, and multiplying that by 12 cans results in a total cost of $9.48.

Long Form Response

11. **Answer:** To find the total number of milkshakes sold at the ice cream shop, we start by using the information that 48 vanilla milkshakes represented 40% of the total sales. We can set up an equation where 40% of the total sales (let's call it x) equals 48. This can be expressed as 0.4 x = 48. To solve for x, we divide both sides of the equation by 0.4, which gives us x = 48 / 0.4. Performing the calculation, we find that x equals 120. Therefore, the total number of milkshakes sold that day was 120.
    Note: correct because it accurately sets up and solves the equation that relates the known quantity of vanilla milkshakes to the total sales, using the percentage relationship to find the total number of milkshakes sold.

12. **Answer:** To determine the number of boxes needed for the remaining muffins, we first need to calculate how many muffins are left after sending some to the hotel. The baker made 232 muffins and sent 190 to the hotel, so we subtract 190 from 232. This can be represented by the equation: $232 - 190 = 42$. Now, we have 42 muffins remaining. If each box can hold 6 muffins, we then divide the remaining muffins by the number of muffins per box, which gives us $42 \div 6 = 7$. Therefore, the baker needs 7 boxes for the remaining muffins.
    Note: correct because it accurately calculates the remaining muffins by subtracting the number sent to the hotel from the total made and then determines the number of boxes needed by dividing the remaining muffins by the capacity of each box.

End of Answer Key